

src/app/BackgroundActivityContext.tsx

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Overview

A new file `src/app/BackgroundActivityContext.tsx` (lines 1-166) introduces a React context for tracking background activities across the app.

Key changes

- **Imports:** `createContext`, `useCallback`, `useContext`, `useMemo`, `useState`, `ReactNode` from React (R1-6).
- **Types:** `BackgroundActivityKind` enum (R10-19) and `BackgroundActivity` interface (R22-30).
- **Helpers:** `sameActivity` comparison (R37-45) and `UpsertPatch` type (R32-35).
- **Contexts:** `BackgroundActivityContext` and `BackgroundActivityActionsContext` (R58-64).
- **Provider:** `BackgroundActivityProvider` (R65-138) manages an array of activities, exposes `upsertActivity`, `removeActivity`, `clearActivities`, and sorts by `startedAt` (R85-96).
- **Hooks:** `useBackgroundActivity`, `useBackgroundActivityOptional`, `useBackgroundActivityActions`, `useBackgroundActivityActionsOptional` (R140-166) with runtime checks (R141-144, R155-160).

Impact

- Centralized activity state reduces duplication; optional hooks allow safe use when the provider may be absent.
- Runtime errors are thrown if hooks are used outside the provider (lines 141-144, 155-160).
- Activities are sorted on every upsert; the list is small, so re-render cost is expected to be low.

Risks & follow-ups

- **Provider omission:** components using the hooks must be wrapped by `BackgroundActivityProvider`; otherwise errors will occur.
- **Sorting side-effects:** verify that ordering by `startedAt` matches UI expectations during concurrent upserts.
- **Optional hooks:** confirm that `useBackgroundActivityOptional` and `useBackgroundActivityActionsOptional` return `null` when the provider is missing.
- Run `npm run lint`, `npm test`, and `npm run build` to ensure the new module compiles and passes tests.